0.1 Unit L and unbounded η: all testing angles

Theorem 1. *Keep the native C1 source with $d = 1$, $x, \sigma = 1/\eta$ positive infinitesimal, $L = x\zeta$ a positive bounded unit, and all original strict gaps and nonnegative event coefficients. The complete parent is a tensor sum of squares at every independent projective angle pair and every real or complex state, including either permitted face $D = 0$ or $\beta = 0$.*

Proof. The source formulas (P1) and opening-safe estimates (BE3)–(BE9) give

$$C/C_0, A/A_0, \alpha/a_0 \sim 1, \qquad C_0 = \frac{1}{2L(1-r)}, \quad A_0 = \frac{1+L(1-r)}{2Lr}, \quad a_0 = \frac{L}{2(1-Lr)}.$$

Choose real $0 < \ell \le L \le U$ and a real $0 < \mu \le \min(1/4, C, A)$. Such a μ exists uniformly through the source openings because $C_0, A_0 \ge 1/(2U)$.

First let u be bounded. Put $c = (1+v^2)^{-1/2}$, $s = vc$, $P = uW$, $T = cZ$. When u, c are both infinitesimal, $w = 1/v$ is infinitesimal and Theorem **??** applies, since $z = 1/L$ is bounded. This includes the literal corner. Otherwise put $(u_0, c_0, s_0) = \mathrm{res}(u, c, s)$ and choose real $M \ge 1$, $\tau > 0$ with $|u_0| \le M$ and $u_0^2 + c_0^2 \ge \tau^2$. The next reserve calculation is evaluated at this real testing center, so its matrices are pure source.

The leading selected quarter-parent in the four raw roots (P, X, Y, T) is

$$S_0 = C_0(P-Y)^2 + A_0(1+u^2)Y^2 + a_0(1+u^2)(sY-T)^2 + K_B/2,$$

$$K_B = X^2 + c^2(P-Y)^2 + s^2Y^2 - 2sPT + 2ucX(sP - sY + T) + u^2s^2X^2 + u^2(sY-T)^2.$$

Its $T = 0$ block is strictly positive. Its determinant is

$$\det G_0 = \frac{(1+u^2)^2(Lc^2 + u^2s^2)\{1+L(1-r)\}^2}{16L^2r(1-r)(1-Lr)} > 0.$$

All source-opening denominators cancel from the cofactor ratio:

$$\mathrm{tr}\, G_0^{-1} = \frac{2(L^2N_2 + LN_1 + N_0)}{(1+t)\{Lq + t(1-q)\}\{1+L(1-r)\}}, \quad t = u^2, \ q = c^2,$$

where, writing $e_r = 1 - r$,

$$N_2 = 1 + r^2 + e_rq\{1 + r(1-q)\} + e_r(2-qr)t + e_r^2(1-q)t^2,$$
$$N_1 = 2e_rq + \{2 - rq(3-q)\}t + e_r(1-q)t^2, \qquad N_0 = 1 + (2-q)t.$$

The numerator is bounded by a real constant depending only on U, M. Moreover $(1+t)\{Lq + t(1-q)\} \ge \min(\ell, 1)(q+t)$. Thus $G_0 \succeq \kappa I$ for some real $\kappa > 0$, uniformly through every source opening.

Set $d_s = |C/C_0 - 1| + |A/A_0 - 1| + |\alpha/a_0 - 1|$ and $f = 1 - d_s$. Retain the exact original event excesses $C - fC_0, A - fA_0, \alpha - fa_0$. The remaining actual form is exactly $fS_0 + (1-f)K_B/2$. The core at the real center has bounded coefficients, whereas d_s is source infinitesimal. Its actual source Gram therefore has a positive real lower bound.

Use the moving event coordinates $e = (P - Y, Y, sY - T, X)$. Both coordinate maps have squared Frobenius norm at most six. Clip C, A, α at a sufficiently large real N, retaining each excess as its whole original event. In these coordinates the core has norm at most a real K, independently of source openings. If the unmodified Gram has reserve $k > 0$, its complementary block remains at least k, its clipped large block is at least $N - K$, and the off-diagonal norm is at most K. Taking $N > K + 2K^2/k + 1$ preserves a positive real reserve, by $2Kab \le (2K^2/k)a^2 + (k/2)b^2$.

The clipped coefficients are bounded source elements. Its entire live Gram, including both state normals, is a polynomial in bounded testing u, c, s in the moving coordinates e. Replacing its testing coefficients by u_0, c_0, s_0 changes the Gram by bounded zero-residue coefficients. The source matrix at that center has positive-definite real residue by the preceding reserve. The separated bounded-Gram lemma applies. Restore all clipped excesses and the complete original m, D, β terms.

It remains to treat unbounded u. Put $p = 1/u$ and use two Bob charts. For $|v| \le 1$ take (W, Y, B, R) with $B = pX + Z$, $R = vX$; the exact core is

$$K_p = B^2 - 2pvBW - 2vBY + (1+p^2)R^2 + 2(1+p^2)RW + (W - pY)^2 + (1+p^2)v^2Y^2.$$

Select only $C = A = \mu$. At $p = 0$ the selected quarter-parent is

$$\mu(1+v^2)(W^2+Y^2) + \tfrac{1}{2}(W+R)^2 + \tfrac{1}{2}(B-vY)^2 \succeq (\mu/3)I.$$

For $|w| = |1/v| \le 1$ use (W, Y, X, H), $H = wZ$, and the exact core

$$\begin{aligned} K_{pw} = {} & H^2 - 2pHW + 2pwHX - 2HY + w^2W^2 + 2wWX - 2pw^2WY \\ & + (1 + p^2 + p^2w^2)(X^2 + Y^2) - 2pwXY. \end{aligned}$$

The corresponding selected value at $p = 0$ is

$$\mu(1+w^2)(W^2+Y^2) + \tfrac{1}{2}(X+wW)^2 + \tfrac{1}{2}(H-Y)^2 \succeq (\mu/3)I.$$

In each chart every remaining coefficient is bounded infinitesimal. The complete selected testing Gram is therefore positive definite and can be factored entirely in the testing field. Restore $C - \mu$, $A - \mu$ and all other original nonnegative event terms. The two charts cover every Bob angle, including their literal endpoints.

All normalizations are positive pure testing factors. The exact sign gauges are $u \mapsto -u$, $(Y, Z) \mapsto (-Y, -Z)$, and $v \mapsto -v$, $(X, Z) \mapsto (-X, -Z)$; they preserve the full parent. Real and imaginary splitting treats complex states. No source coefficient D or β was divided by or capped. $\square$